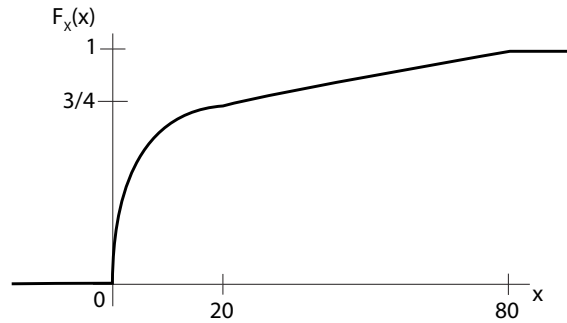


Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE 2500: Engineering Probability, Spring 2024
Homework #4 Solutions

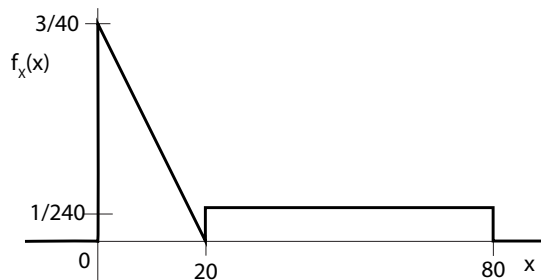
1. (a) The CDF is illustrated below. The first region is part of a quadratic function, and the second region is a straight line segment. We can see that most of the probability mass is before 20 minutes (which makes sense since you would notice a missing calculator quickly).



- (b) Since $x = 10$ is in the first (quadratic) piece, we can easily compute that $P(X < 10) = F_X(10) = \frac{3}{1600} (400 - (10 - 20)^2) = \frac{9}{16} = 0.5625$.
- (c) To compute $P(X \in [15, 40])$, note that 15 and 40 are in different pieces of the CDF. The answer is $F_X(40) - F_X(15) = \frac{5}{6} - \frac{45}{64} = \frac{25}{192} = 0.1302$.
- (d) To compute the PDF we take the derivative (in this case, piece by piece):

$$f_X(x) = \begin{cases} -\frac{3}{800}(x - 20) & x \in [0, 20] \\ \frac{1}{240} & x \in [20, 80] \\ 0 & \text{otherwise} \end{cases}$$

The PDF is sketched below; remember to show that it's 0 outside of the key region! In this way we can see that 1/4 of the probability is in the region $X \in [20, 80]$ and 3/4 of the probability is in the region $X \in [0, 20]$.

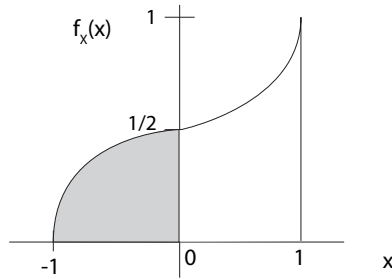


2. (a) We need the PDF to integrate to 1, i.e.,

$$\begin{aligned}
 \int_{x=-\infty}^{\infty} f_X(x) dx &= \int_{x=-1}^1 c(x^3 + 1) dx \\
 &= \left. \frac{1}{4}cx^4 + cx \right|_{x=-1}^{x=1} \\
 &= c\left(\frac{5}{4} + \frac{3}{4}\right) \\
 &= 2c \\
 &= 1
 \end{aligned}$$

Thus we must have $c = \frac{1}{2}$.

- (b) The PDF is illustrated below. It looks like a cubic function that turns on at -1 and off at 1, shifted upwards.



- (c) $P(X < 0)$ corresponds to the shaded region in the figure above, which we can compute as

$$\begin{aligned}
 \int_{x=-1}^0 \frac{1}{2}(x^3 + 1) dx \\
 &= \left. \frac{1}{8}x^4 + \frac{1}{2}x \right|_{x=-1}^{x=0} \\
 &= 0 - \left(\frac{3}{8}\right) \\
 &= \frac{3}{8}
 \end{aligned}$$

- (d) To get the CDF from the PDF, we need to integrate. Below $x = -1$ the CDF is 0 and above $x = 1$ the CDF is 1, so we only need to figure out the “middle piece”:

$$\begin{aligned}
 F_X(x) &= \int_{y=-\infty}^x f_X(y) dy \\
 &= \int_{y=-1}^x \frac{1}{2}(y^3 + 1) dy \\
 &= \left. \frac{1}{8}y^4 + \frac{1}{2}y \right|_{y=-1}^{y=x} \\
 &= \frac{1}{8}x^4 + \frac{1}{2}x + \frac{3}{8}
 \end{aligned}$$

So the CDF overall is:

$$F_X(x) = \begin{cases} 0 & x < -1 \\ \frac{1}{8}x^4 + \frac{1}{2}x + \frac{3}{8} & x \in [-1, 1] \\ 1 & x > 1 \end{cases}$$

3. The exponential PDF with $\lambda = \frac{1}{3}$ is

$$f_X(x) = \frac{1}{3}e^{-\frac{1}{3}x}u(x)$$

where $u(x)$ denotes the unit step function that turns on at 0.

(a) No violations in 2 hours means $X > 2$ so we compute

$$\begin{aligned} P(X > 2) &= \int_{x=2}^{\infty} f_X(x) dx \\ &= \int_{x=2}^{\infty} \frac{1}{3}e^{-\frac{1}{3}x} dx \\ &= -e^{-\frac{1}{3}x} \Big|_{x=2}^{x=\infty} \\ &= e^{-\frac{2}{3}} \\ &= 0.5134 \end{aligned}$$

(b) Recall that if the arrival times X between events are exponentially distributed, the number of events N in a given interval is a Poisson random variable. Here, $\lambda = \frac{1}{3}$ and $t = 6$, so the probability is

$$\begin{aligned} P(N = 3) &= e^{-\lambda t} \frac{(\lambda t)^3}{3!} \\ &= e^{-2} \frac{2^3}{3!} \\ &= 0.6846 \end{aligned}$$

(c) Now we want to compute λ so that X is equally likely to be on either side of 1, which is the same thing as solving

$$\int_{x=1}^{\infty} \lambda e^{-\lambda x} = \frac{1}{2}$$

Using the same logic as in part (a), this reduces to

$$e^{-\lambda} = \frac{1}{2}$$

or $\lambda = \log 2 = 0.6931$.

4. In this problem, X is a Gaussian random variable with mean 6 and standard deviation 2. We will also use Z , a Gaussian random variable with mean 0 and standard deviation 1. The idea is to convert every statement about X into a statement about Z so we can use the Q tables for the CDF of Z .

(a)

$$\begin{aligned}P(X > 5) &= P\left(\frac{X-6}{2} > \frac{5-6}{2}\right) \\&= P(Z > -0.5) \\&= 1 - Q(0.5) \\&= 1 - 0.3085 \\&= 0.6915\end{aligned}$$

(b)

$$\begin{aligned}P(|X-6| > 0.6) &= P\left(\frac{|X-6|}{2} > \frac{0.6}{2}\right) \\&= P(|Z| > 0.3) \\&= 2Q(0.3) \\&= 2(0.3821) \\&= 0.7642\end{aligned}$$

(c)

$$\begin{aligned}P(X \in [3, 7.8]) &= P\left(\frac{X-6}{2} \in \left[\frac{3-6}{2}, \frac{7.8-6}{2}\right]\right) \\&= P(Z \in [-1.5, 0.9]) \\&= 1 - Q(1.5) - Q(0.9) \\&= 1 - 0.0668 - 0.1841 \\&= 0.7491\end{aligned}$$

5. Here, Y is a Gaussian random variable with mean 30 and standard deviation 5. We again must convert every statement about Y into a statement about a standard Gaussian Z , but this time we're using the inverse of the Q tables.

(a)

$$P(Y > a) = P\left(Z > \frac{a-30}{5}\right)$$

This means $Q\left(\frac{a-30}{5}\right) = 0.2$. Eyeballing from the table the value z such that $Q(z) = 0.2$ is close to 0.8 (WolframAlpha tells us it's more like 0.8416). Thus

$$a = 5Q^{-1}(0.2) + 30 = 5(0.8416) + 30 = 34.208$$

(b)

$$P(|Y - 30| < b) = P\left(|Z| < \frac{b}{5}\right)$$

This means $Q\left(\frac{b}{5}\right) = 0.3$. That is, we want 0.3 of the total probability within the range $Z \pm \frac{b}{5}$, so there is 0.35 of the probability left in each tail. Eyeballing from the table the value z such that $Q(z) = 0.35$ is close to 0.4 (WolframAlpha tells us it's 0.3853). Thus

$$b = 5Q^{-1}(0.3) = 5(0.3853) = 1.9265$$

(c) This is a conditional probability problem.

$$\begin{aligned} P(X > 38 \mid X > 32) &= \frac{P(X > 38 \cap X > 32)}{P(X > 32)} \\ &= \frac{P(X > 38)}{P(X > 32)} \\ &= \frac{P(Z > 1.6)}{P(Z > 0.4)} \\ &= \frac{Q(1.6)}{Q(0.4)} \\ &= \frac{0.0548}{0.3446} \\ &= 0.1590 \end{aligned}$$